

Deepak Mallampati

AI ENGINEER - ENTERPRISE AI AGENTS, RAG APPLIED AI ENGINEER - AGENTIC WORKFLOWS & AI TOOLINGAMP; AZURE

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US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

AI Engineer with 4+ years building production GenAI and LLM systems, including 2+ years shipping agentic and RAG applications in regulated, enterprise-scale environments (100,000+ daily requests, 99.9% uptime). Builds AI agents and multi-agent systems, MCP-style tool integrations, RAG pipelines, orchestration patterns, and automated evaluation loops, with advanced Python and RESTful API design. Cloud experience spans Azure (OpenAI, AI Services, Content Safety, DevOps) and AWS, with CI/CD and responsible-AI controls throughout - a direct fit for enterprise AI delivery in the Microsoft ecosystem.

CORE COMPETENCIES

AI Agents & Multi-Agent Systems • MCP Servers / Tool Integration • RAG • Orchestration Patterns • Azure AI • Prompt Engineering & Evals • RESTful APIs • Python • Responsible AI & CI/CD

Generative AI & LLMs: LangChain/LangGraph, RAG architecture, multi-agent systems, prompt engineering, tool-use/function-calling, GPT-4o/GPT-5, Claude, Gemini, LLaMA

Vector Search & Retrieval: FAISS, Pinecone, Weaviate, semantic & hybrid search, cross-encoder re-ranking, document chunking, semantic caching

Cloud & MLOps: Azure (OpenAI, AI Services, Content Safety, Azure DevOps, AI Foundry-adjacent), AWS (Bedrock, SageMaker, Lambda), Docker, Kubernetes, Terraform, MLflow, CI/CD

Security & Compliance: prompt-injection mitigation, PII masking/redaction, RBAC, OAuth 2.0/JWT, audit logging, zero-trust, HIPAA/SOC 2/PCI DSS

Languages & Tools: Python, JavaScript/TypeScript, SQL, FastAPI, Git; PostgreSQL, MongoDB, Redis; Docker, Kubernetes

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing objectives into context-scoped subagents executing autonomously - **42%** token reduction at sustained accuracy across multi-step workflows.
- Authored and iterated **agentic harnesses and AI tooling** with tool-use/function-calling and model routing, plus fault-tolerant automation infrastructure (scheduled batch orchestration, exponential-backoff retry) that cut failed production runs by **60%**.
- Engineered **RAG pipelines** over large healthcare corpora (dense retrieval + cross-encoder re-ranking), lifting answer relevance and materially reducing hallucination on domain-specific queries.
- Built privacy-preserving workflows (K-anonymity, L-diversity, differential privacy) and an **LLM evaluation framework** (RAGAS, BERTScore) with latency/accuracy dashboards.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated GPT-4o, Claude, and Gemini, improving task success by **27%**; engineered content-safety controls reducing unsafe outputs by **42%**.
- Built **12 APIs and microservices**, cutting p95 latency 1.5s → **800ms** and supporting **100,000+** requests daily across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline on clinical data; reduced re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned Qwen3 via 4-bit QLoRA on an H100; authored an IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Built regression/classification models for failure prediction, anomaly detection, and forecasting; fine-tuned BERT/RoBERTa for NER reaching **89% F1**.

PROJECTS

career-ops

An autonomous, agentic AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - internal automation and ongoing experimentation with emerging AI in one shipped system.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - single-action inline tooling built end to end.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0